SEED

First Name	Last Name	email

0. Ice melting

The Norwegian Ice Block Expedition of 1959 drove a lorry with three metric tons of ice from the Polar Circle to a hospital in Africa at the Equator – with no refrigeration – using only glass fibre for insulation. Starting with a mass of 3050 kg of ice, what mass remained at the end of the journey? (m)

5%ile	50%ile	95%ile

1. Major active faults in Japan

In its 1999 report, The Promotion of Earthquake Research: Basic Comprehensive Policy for the Promotion of Earthquake Observation, Measurement, Surveys and Research, how many “major active faults” did the Japanese Government identify? (n°)

5%ile	50%ile	95%ile

2. NE Japan regional strain rate

The Centennial Geodetic Analysis Model for northeast Japan combined historical and modern geodetic observations spanning 1894–2009. What central value did the model estimate for the steady maximum shear-strain rate, expressed in nanostrain/yr? [Source: Meneses-Gutierrez, Segall & Sagiya, 2022] (nanostrain/yr)

5%ile	50%ile	95%ile

3. 2011 Tohoku earthquake coseismic subsidence

What was the maximum magnitude of coseismic subsidence recorded by the GNSS/GEONET network along the Pacific coast during the 2011 Tohoku-oki earthquake, in cm? [Source: Nishimura, Munekane & Yarai, 2011] (cm)

5%ile	50%ile	95%ile

4. ScS-associated permanent displacement

Several minutes after the 2011 Tohoku-oki mainshock, ScS waves reflected from the core–mantle boundary produced transient motion at nearly all GNSS stations in Japan. Superimposed on this motion was a permanent, step-like eastward displacement, with a reported maximum of 6 mm. For the Hokkaido stations, what was the estimated mean permanent eastward displacement, expressed as a percentage of 6 mm? [Source: Park, Kanamori & Rivera, 2026] (%)

5%ile	50%ile	95%ile

5. Uplift rate – upper endpoint

For the Shizunai area on the Hidaka coast of southern Hokkaido, a range of long-term late-Quaternary uplift rates was estimated from adjusted relative sea-level elevations of MIS 5e marine terraces (130–116 ka), with peak sea level at approximately 124 ka. The lower endpoint of the reported range is 0.25 mm/yr. What is the corresponding upper endpoint, in mm/yr? [Nakanishi & Ashi, 2022] (mm/yr)

5%ile	50%ile	95%ile

6. Deformation budget fraction

According to Kato et al. (2004), using the long-term whole-system shortening estimate adopted from Ito (2002) and the Amurian–Okhotsk convergence rate from the plate-motion model used in that study, what fraction of regional plate convergence was inferred to be accommodated by shortening across the entire Hidaka thrust system in central Hokkaido? Express your quantiles as decimal fractions rather than percentages. (decimal fraction)

5%ile	50%ile	95%ile